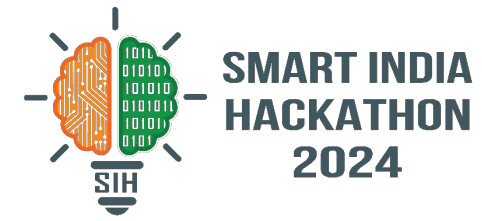
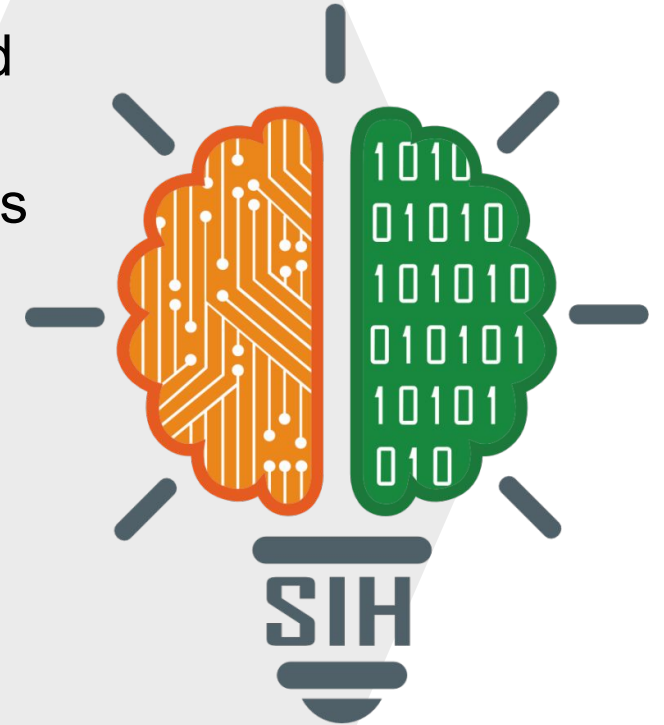


SMART INDIA HACKATHON 2024



- **Problem Statement ID –1647**
- **Problem Statement Title-** Development of AI-ML based models for predicting prices of agri-horticultural commodities such as pulses and vegetable (onion, potato, onion)
- **Theme-**Agriculture, FoodTech & Rural Development
- **PS Category- Software**
- **Team ID-13930**
- **Team Name (Registered on portal)-CodeTemplars**

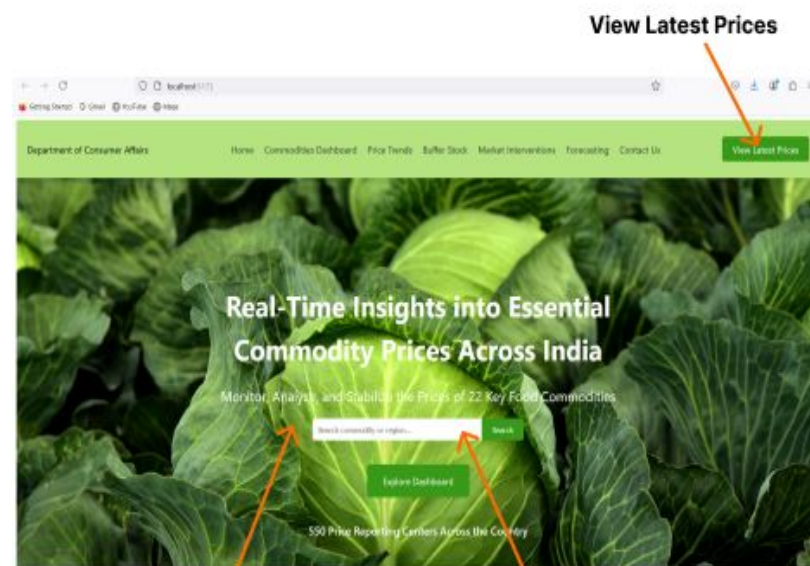
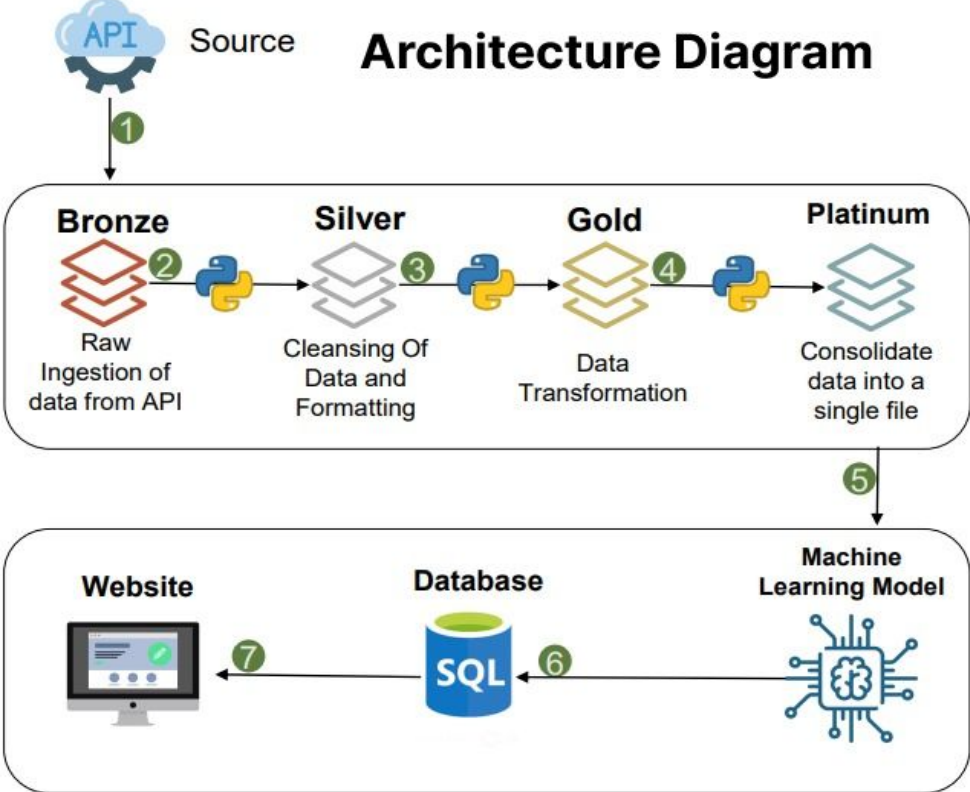


we have created a **Data Monitoring and Forecasting System** using **ARIMA** model that collects daily prices, performs trend analysis, forecasts future prices, and provides strategic insights for market interventions.

- Our approach involve creating a **web-based solution** for providing Real-Time data and Price-Predictions that help prevent extreme price fluctuations, enabling better decision-making for both farmers and consumers.
- For information our platform sends price and buffer alerts notification like “only x left ” also provide **recommendations for releasing buffer stock**.
- Visualise and compare data effortlessly with **graphical representation** like pictorial displays, bar graphs, and line graphs.
- Stay informed with our comprehensive **price prediction model**.
- While many price monitoring systems focus solely on historical price data, this solution incorporates external variables like crop sowing reports and market intelligence, making the forecasts more comprehensive.

Website github link - [Click here](#)

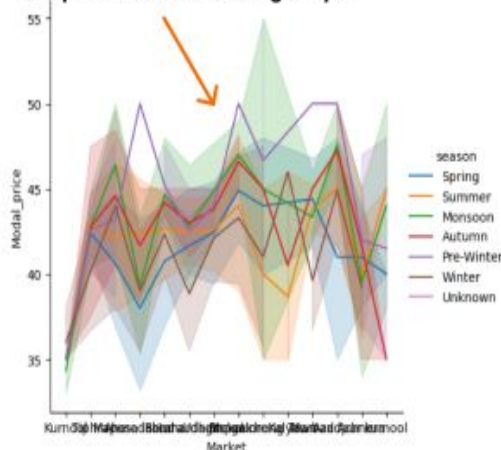
Machine learning Model link- [Click here](#)



Easy to use UI

Search for commodity or region

Comparison of Data using Graphs



Real time Buffer Stock Management



Price Forecasting Graphs



Technologies used- Python, React, SQL, Java script Framework and Libraries(Pandas & NumPy,TensorFlow) etc.

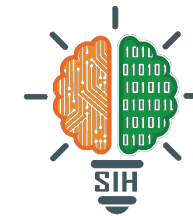
FEASIBILITY AND VIABILITY

- The solution uses readily available technologies (**APIs, ARIMA, cloud services**) and is scalable for future needs, ensuring smooth integration and future-proofing.
- Easy integration for reporting centers with minimal training; secure **cloud-based infrastructure** ensures efficient data handling and protection.
- Moderate development costs with **high ROI** through improved interventions; risks like data quality and adoption can be managed with automated checks and training.
- Ensuring consistent data quality and smooth integration from various sources requires automated checks and thorough testing and it can be managed by Implement automated validation and conduct thorough testing to ensure consistent data quality and smooth integration.
- Overcoming resistance to new technology and ensuring data security necessitates effective training and strong encryption measures it can be managed by Provide comprehensive training and support for users, and employ robust encryption and access controls to address resistance and ensure **data security**.

IMPACT AND BENEFITS

- **For Consumers:** Improved price stability for essential commodities will help reduce the burden of sudden price hikes, ensuring better affordability and access to basic food items.
- **For Farmers and Producers:** Predictable pricing reduces uncertainty in market conditions, enabling more strategic planning for crop production and supply, contributing to overall market stability.
- **For Decision-Makers:** Real-time insights and accurate forecasts enable timely market interventions and better stock management, maintain supply chain .
- Our Website provide benefits by providing Accurate Price Forecasting , **allow farmers to make better decisions** regarding when to sell their produce and when to store it, leading to potentially higher income levels by capitalizing on peak market prices.
- **Help to Accurate Price Forecasting** and also provide **Improved Market Transparency** between large market players and smallholder farmers.

RESEARCH AND REFERENCES



- Study historical data and seasonal trends of essential commodities to understand price fluctuations and factors influencing market behavior.
- Explore ARIMA and other predictive models used in commodity price forecasting to evaluate their effectiveness and adaptability for this solution.
- Investigate best practices for real-time data collection and integration from diverse sources to ensure accuracy, reliability, and scalability.
- For Research and Information AI_Russell_Norvig Book .
- For model Training Dataset: <https://www.kaggle.com/datasets> .
- Our ML Dataset [link](#)